

TRANSPARENT STAYS

BUSINESS INTELLIGENCE AND BIG DATA
ANALYTICS

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DOCUMENT STRUCTURE

The dataset consists of several JSON files where each file includes a collection of hotel reviews.

Fields:

- **Ratings:** numeric rating from 1 to 5 for seven categories - Service, Cleanliness, Value, Sleep Quality, Rooms, Location
- **AuthorLocation:** the city where the review was posted
- **Title:** title of the review
- **Author:** person who wrote the review
- **ReviewID:** unique ID for each review
- **Content:** the review text.
- **Date:** date of the review



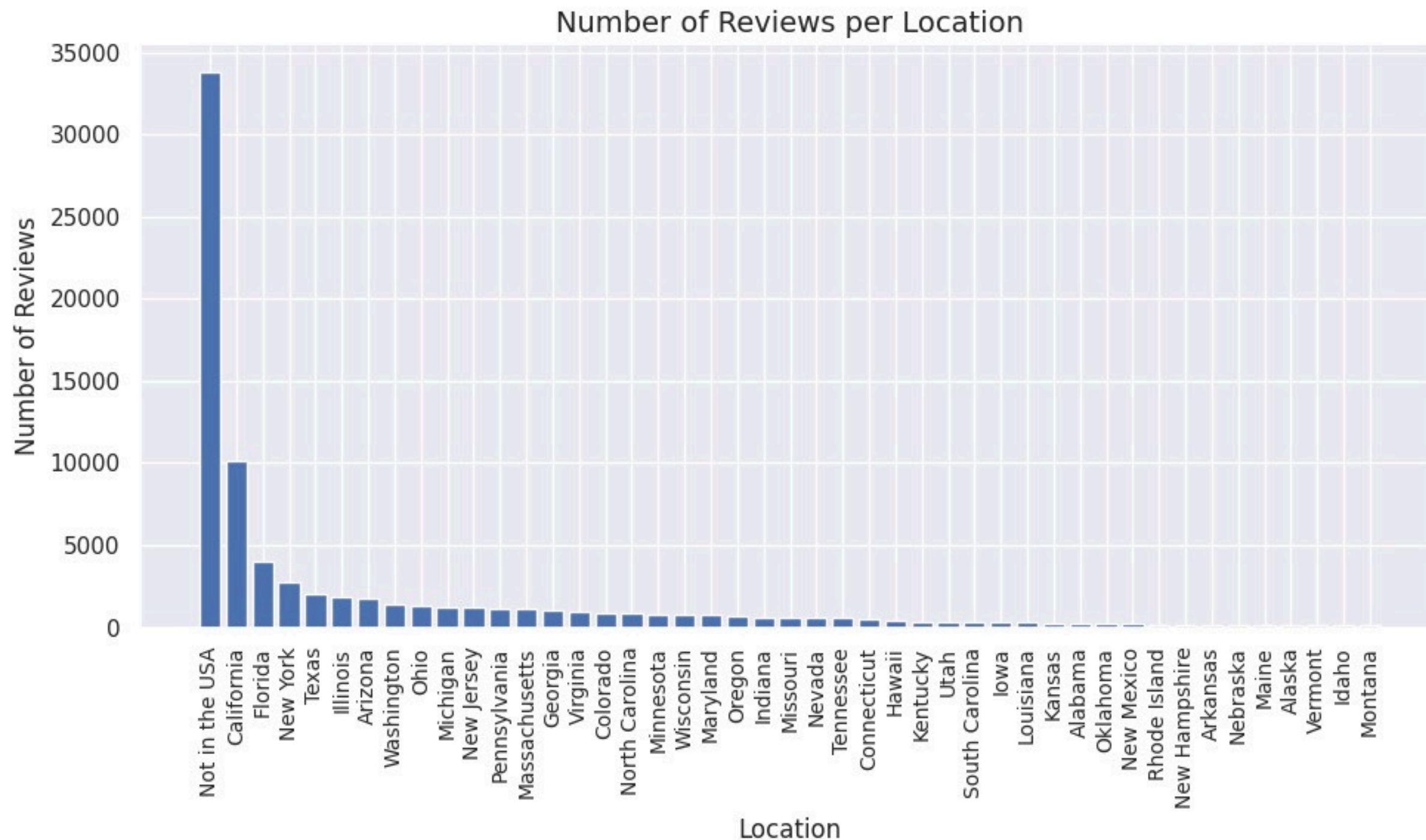
```
_id: ObjectId('67601907ffa73e5fd90634c0')
Ratings : Object
  Service : "5"
  Cleanliness : "5"
  Overall : "5.0"
  Value : "5"
  Sleep Quality : "5"
  Rooms : "5"
  Location : "5"
AuthorLocation : "Boise, ID"
Title : "“Great hotel located minutes from airport.”"
Author : "Dacougr32"
ReviewID : "UR126874246"
Content : "Great little motel located on Marina 5 min from Airport. Point Loma Se..."
Date : "March 28, 2012"
```

AVAILABLE LOCATIONS

We did some pre-processing:

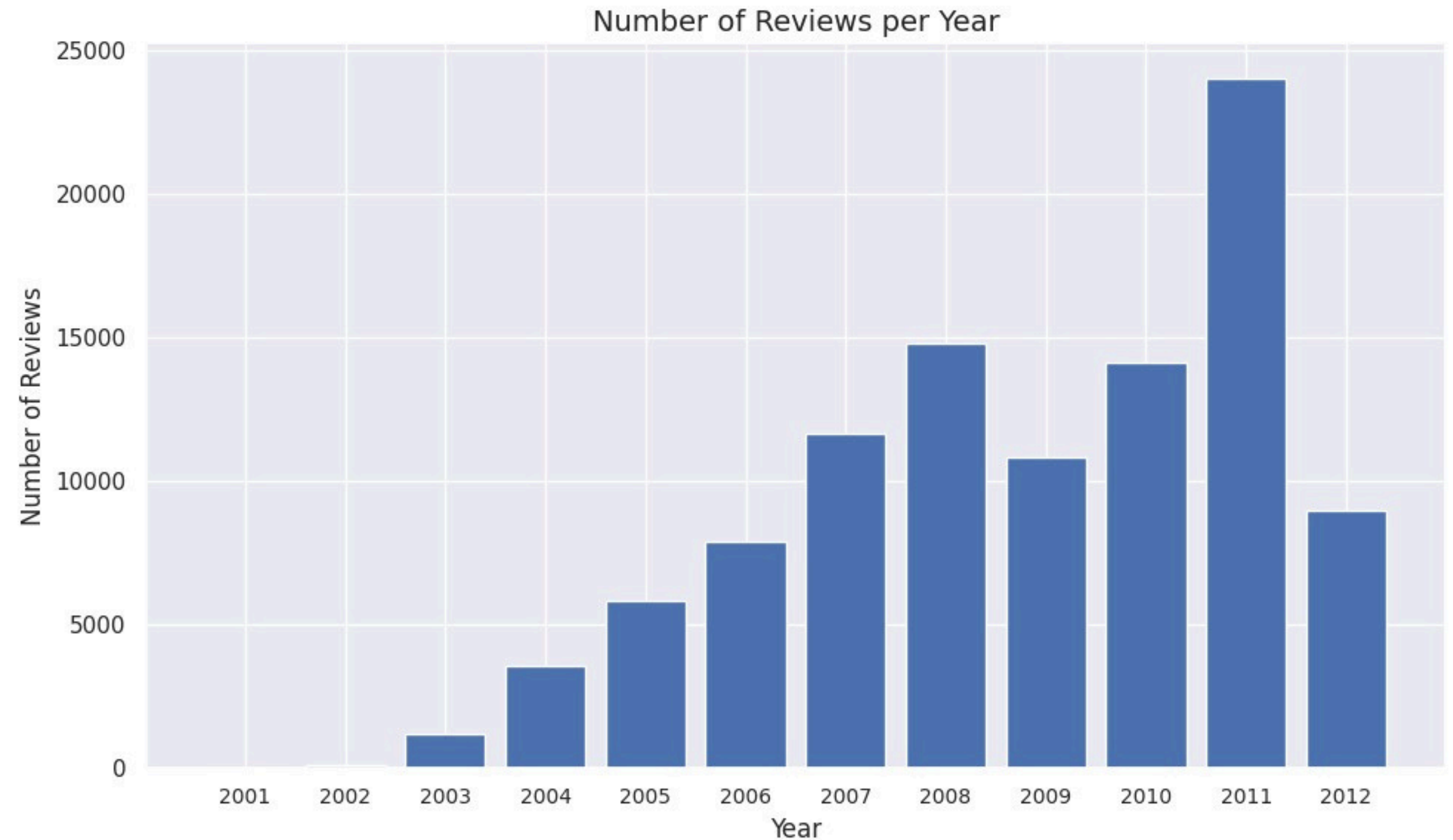
- Normalization of the location
- Division of the location city and state
- Identify if the word matches a valid U.S. state name
- Counting of reviews by location
- Retrieval of locations with the highest number of reviews

We then kept the first three USA locations with the highest amount of reviews.



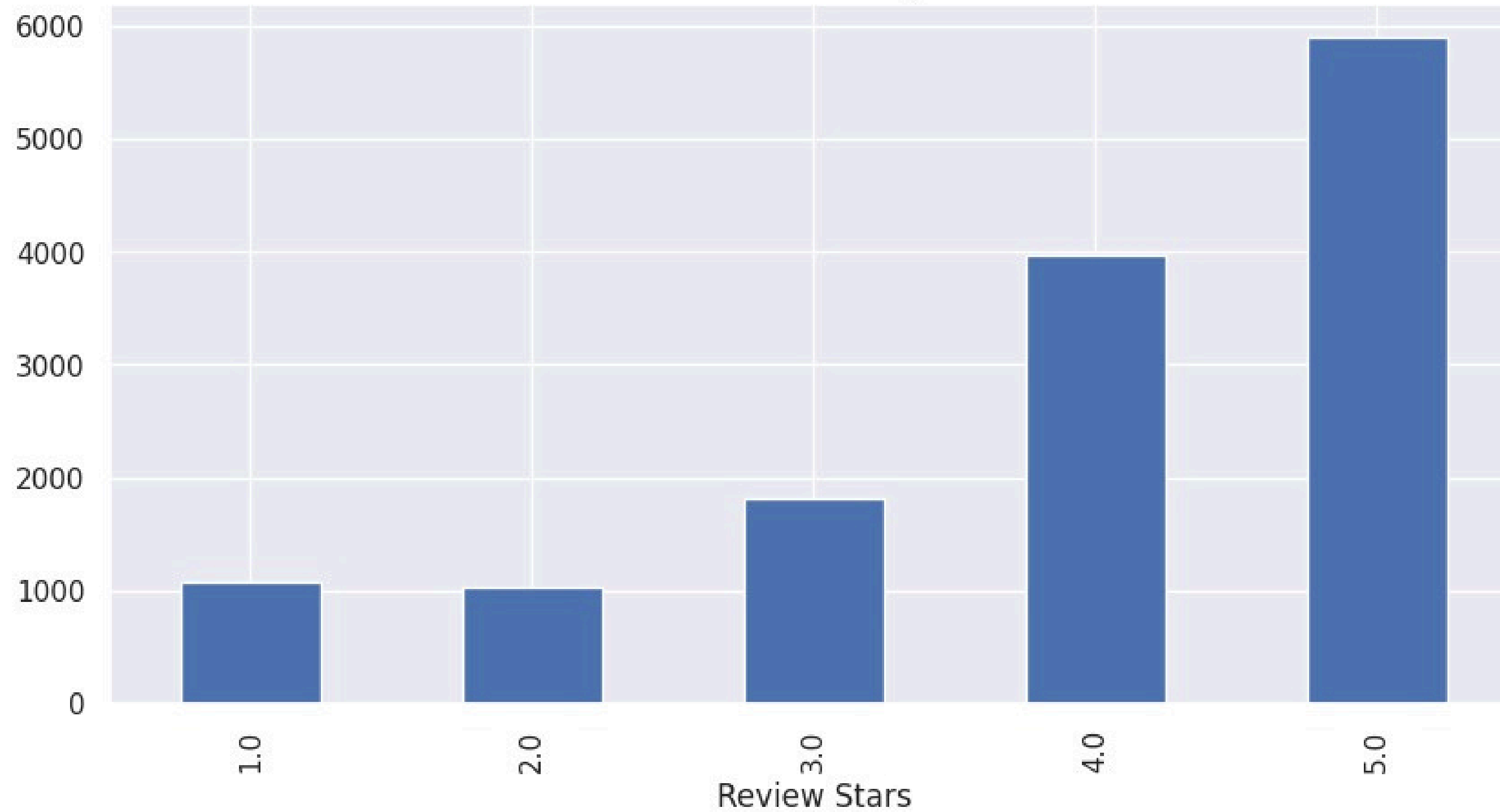
YEARS

- First review's date:
 - August 1, 2001
- Last review's date:
 - September 9, 2012
- We retained reviews from 5 more recent years:
 - from 2008 to 2012



RATING

Count of Reviews by Stars

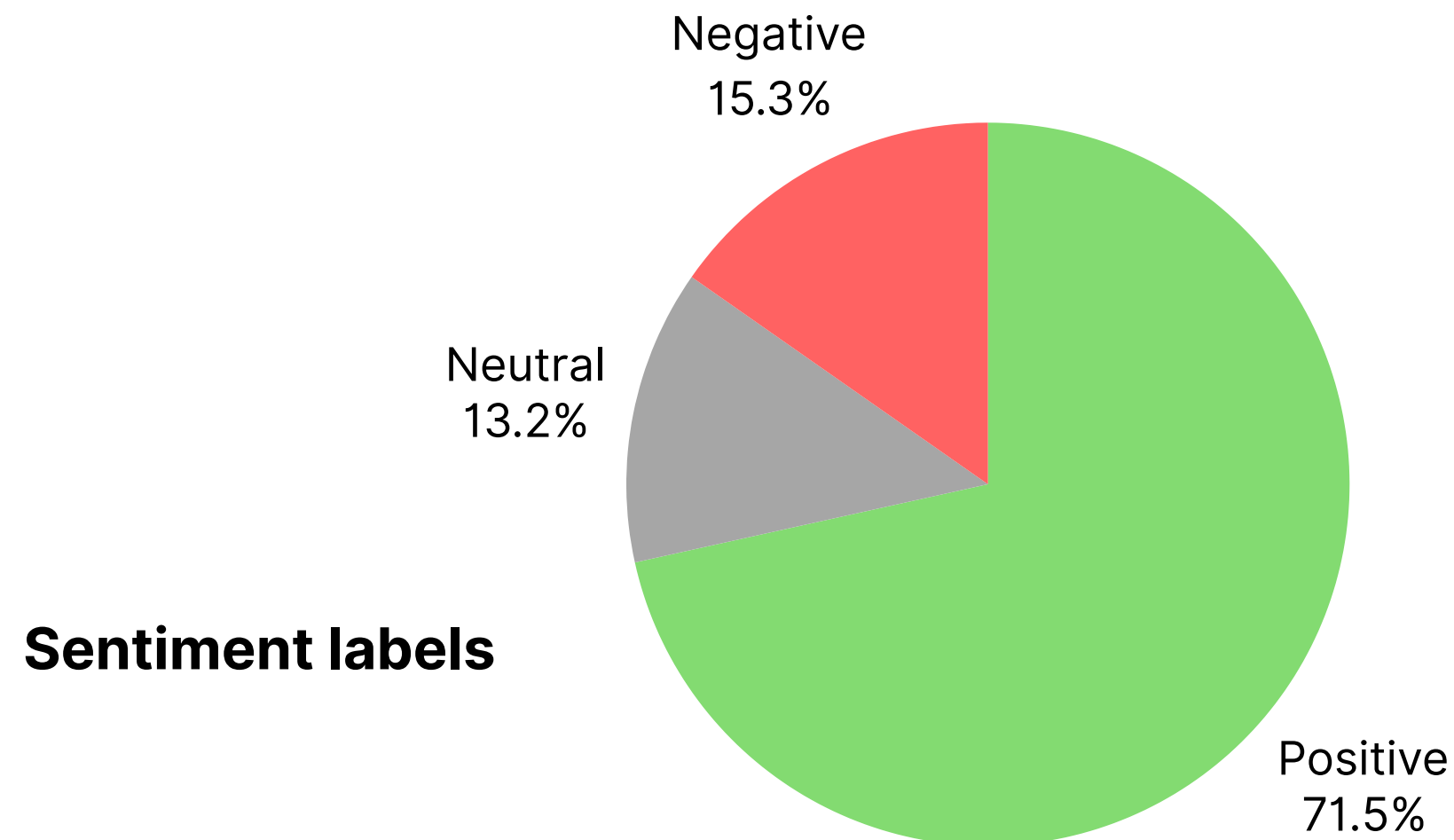


- We considered only the 'overall' rating
- Prevalence of positive reviews

PRE-PROCESSING

- Creation of binary sentiment labels, excluding neutral reviews:
 - if rating ≥ 4 : positive
 - if rating ≤ 2 : negative
- Class imbalance:
 - down-sampling
- Total of 4325 reviews

- Removal of:
 - URL
 - # and @
 - punctuation
- Tokenization
- Stop words removal
- Lemmatization
- Stemming

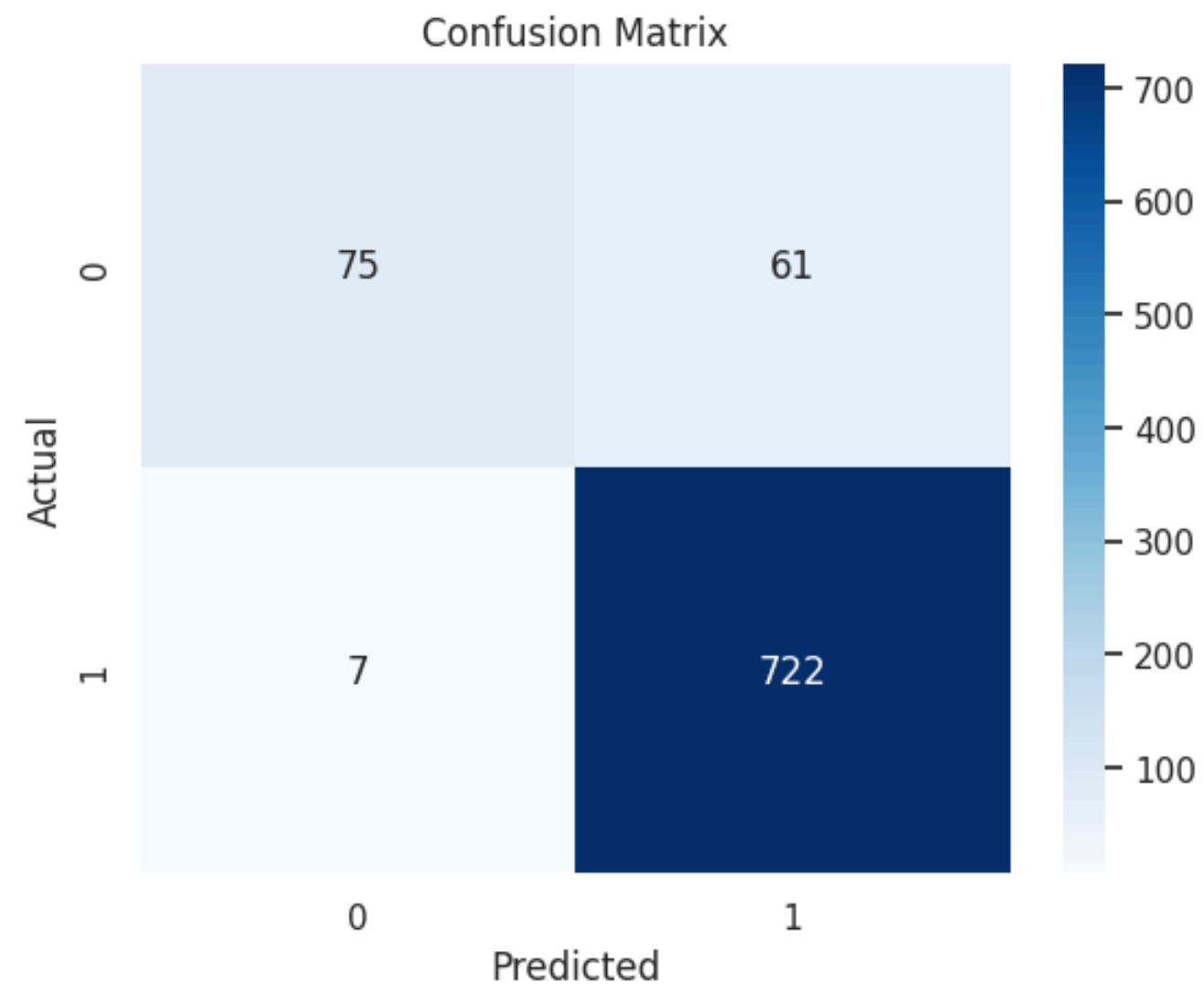


LEXICON-BASED MODELS

VADER

Lexicon-based sentiment analysis tool for the computation of a polarity score.

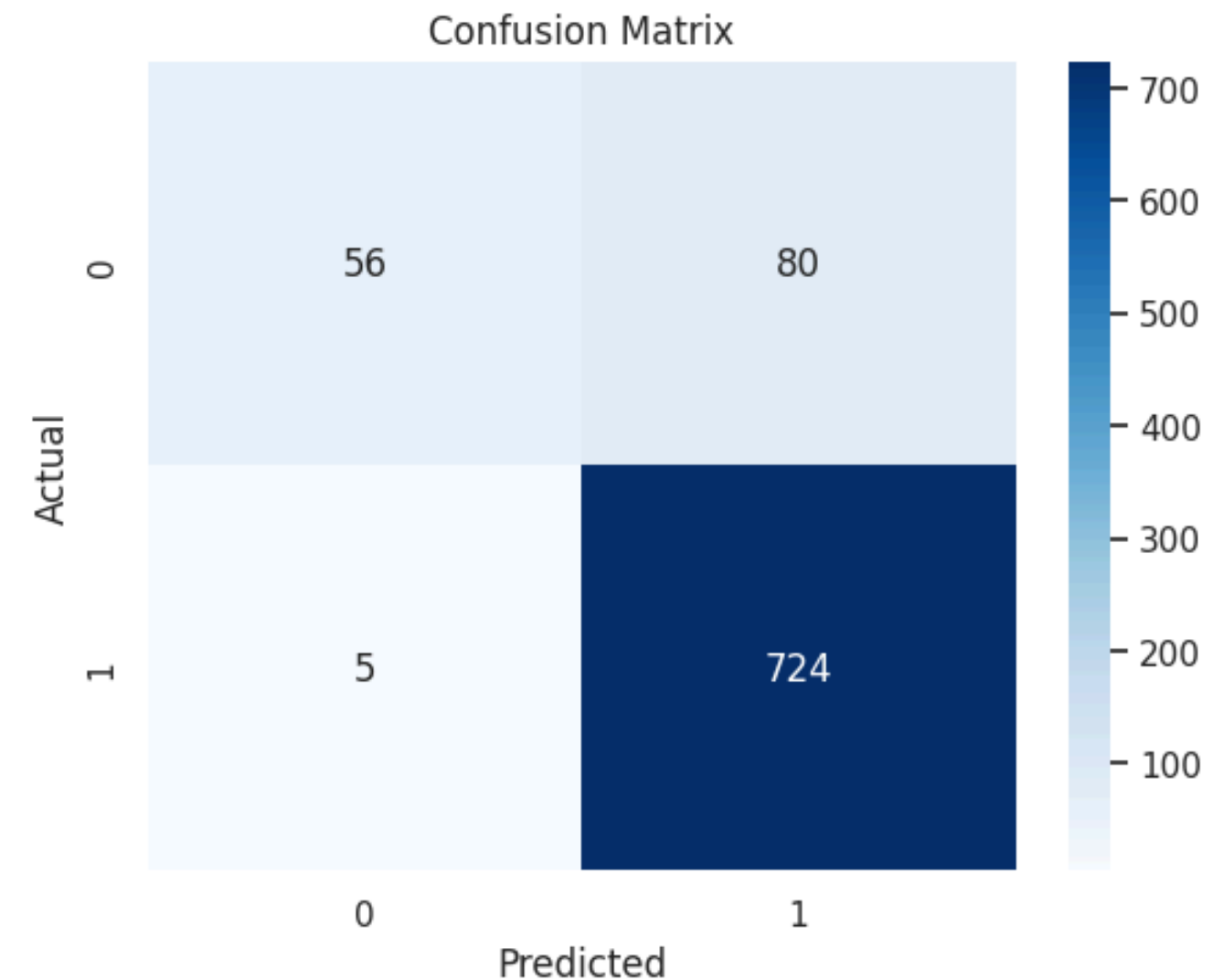
- Specificity: 0.55
- Sensitivity: 0.99



Textblob

Lexicon-based approach which returns a polarity score.

- Specificity: 0.41
- Sensitivity: 0.99

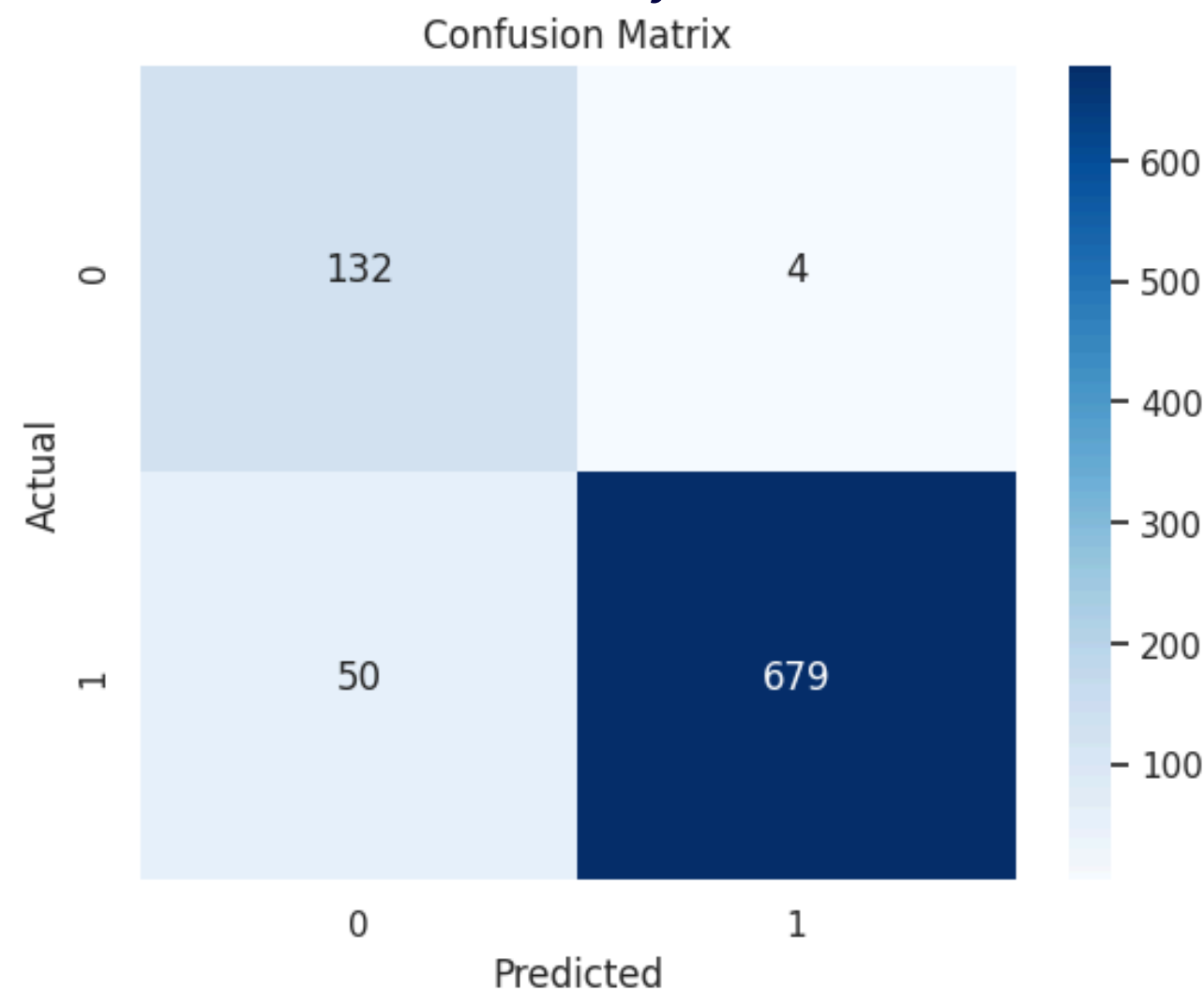


PRE-TRAINED MODELS

Flair

NLP library which provides a pre-trained model for sentiment analysis. Flair is able to extract contextual information from sentences.

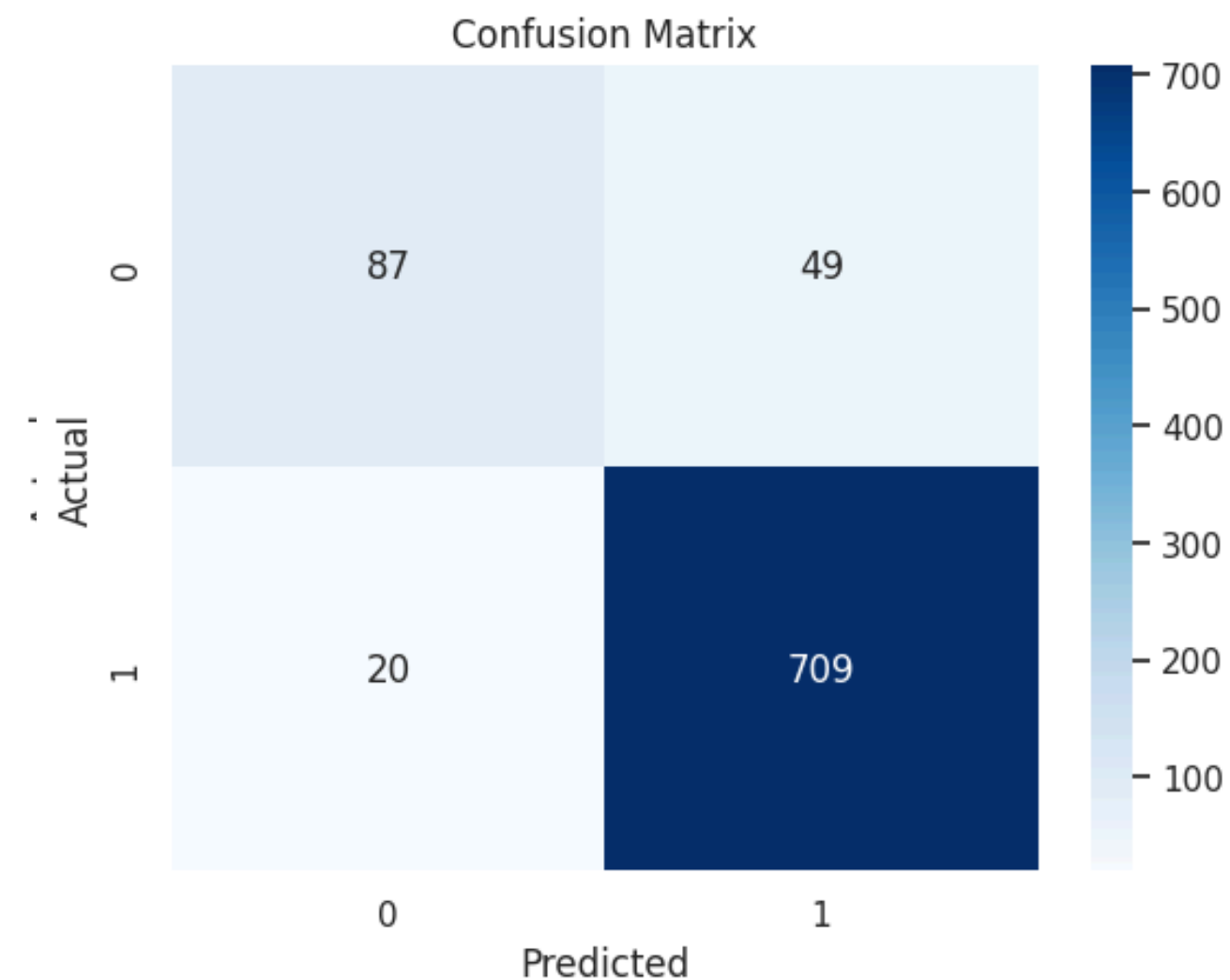
- Specificity: 0.97
- Sensitivity: 0.93



BERT

BERT is a state-of-the-art pre-trained model introduced by Google. It learns information from both the left and the right side of a token's context during the training phase.

- Specificity: 0.64
- Sensitivity: 0.97

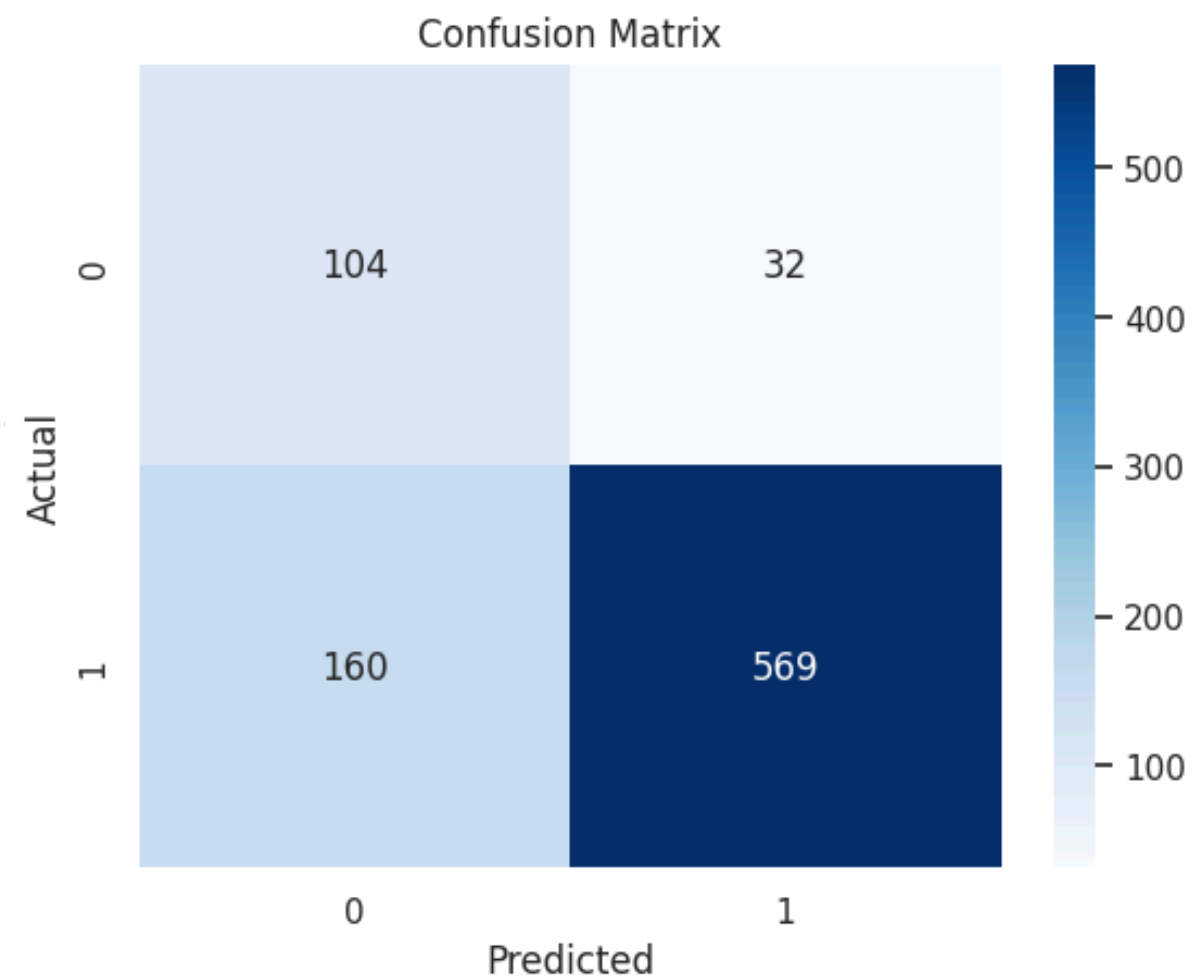


RANDOM FOREST

Word2Vec Embedding

It captures information about the meaning of the word based on the surrounding words

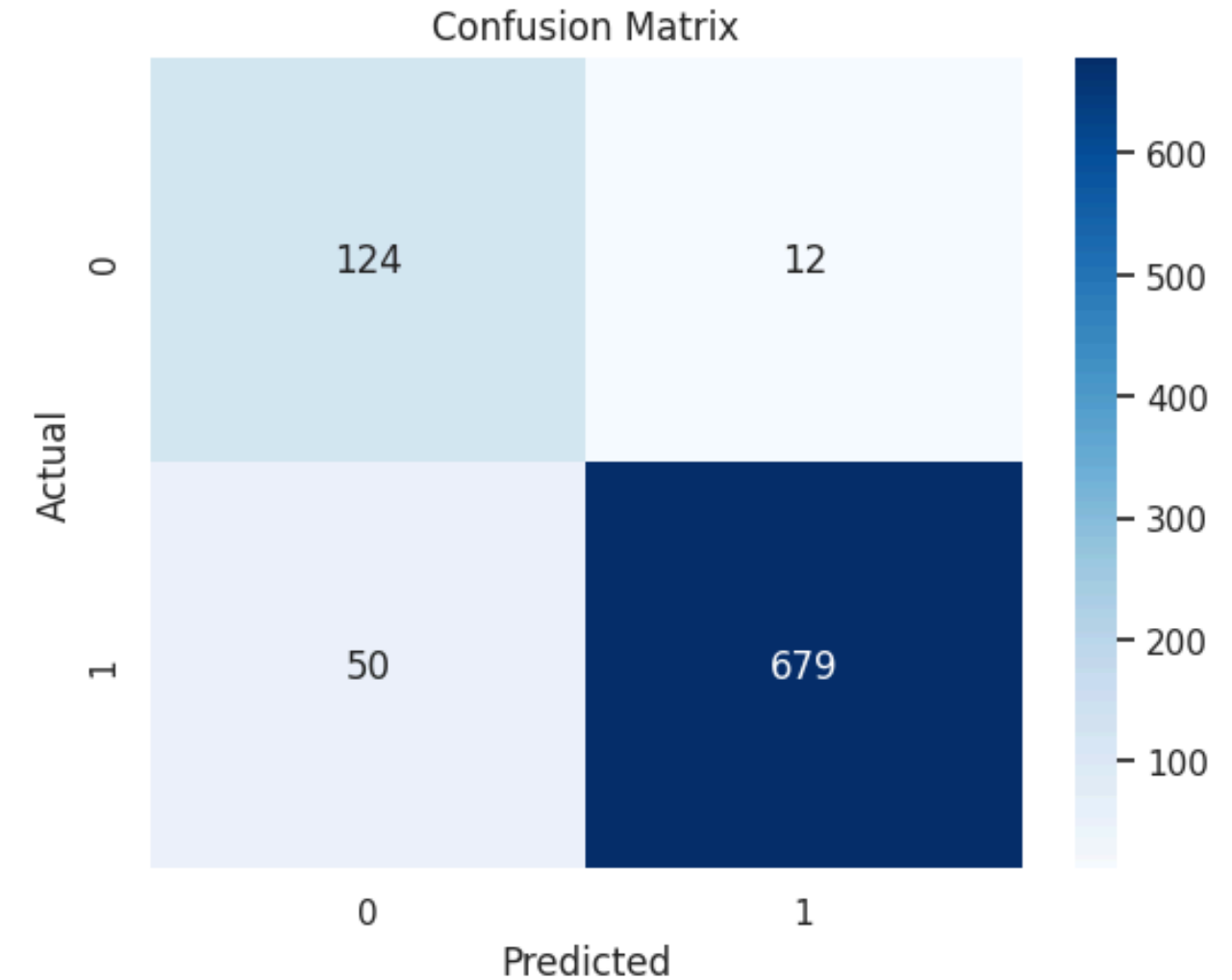
- Specificity: 0.76
- Sensitivity: 0.78



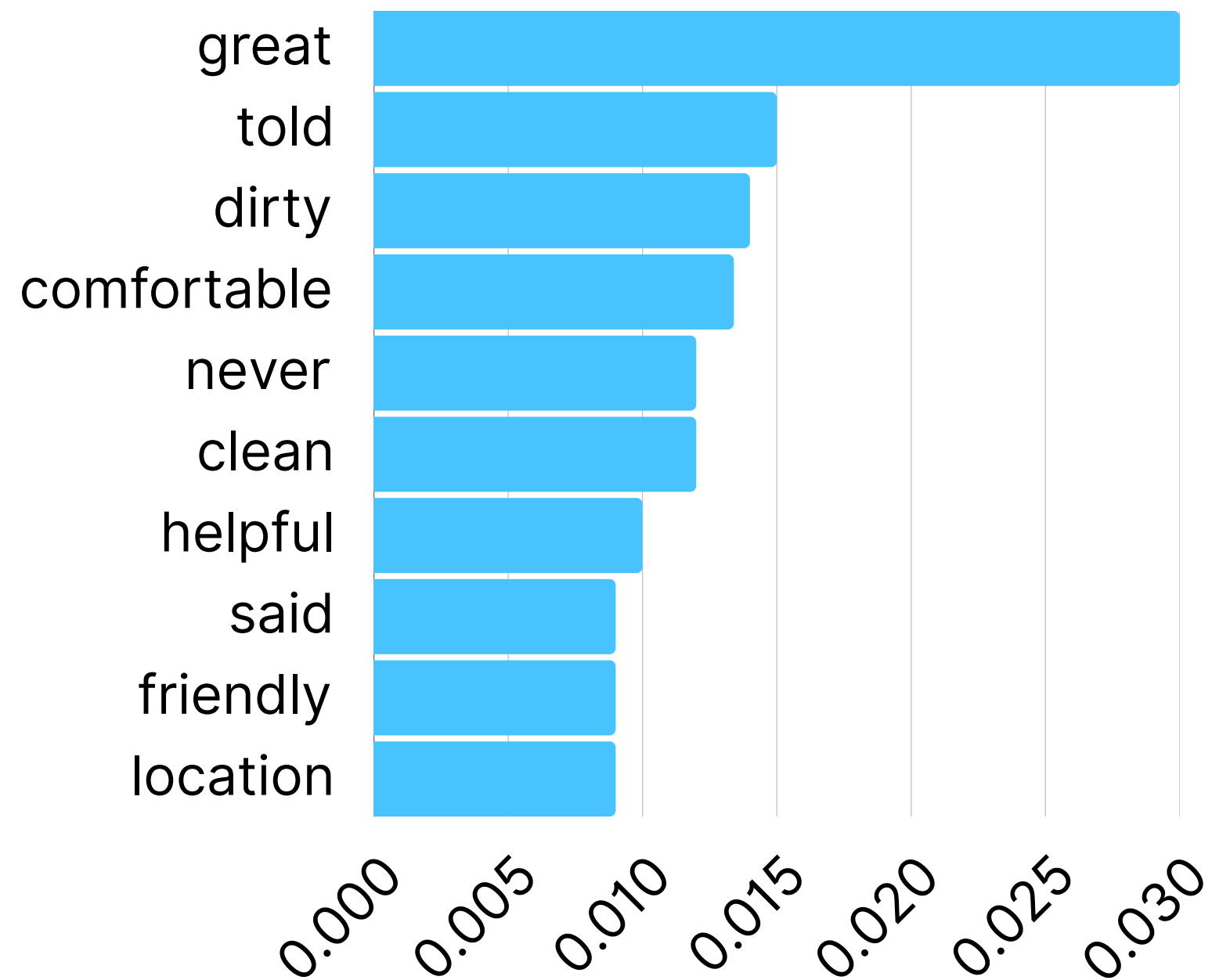
TF-IDF Embedding

It evaluates the importance of a word in a document relative to a collection of documents combining Term Frequency and Inverse Document Frequency

- Specificity: 0.91
- Sensitivity: 0.93



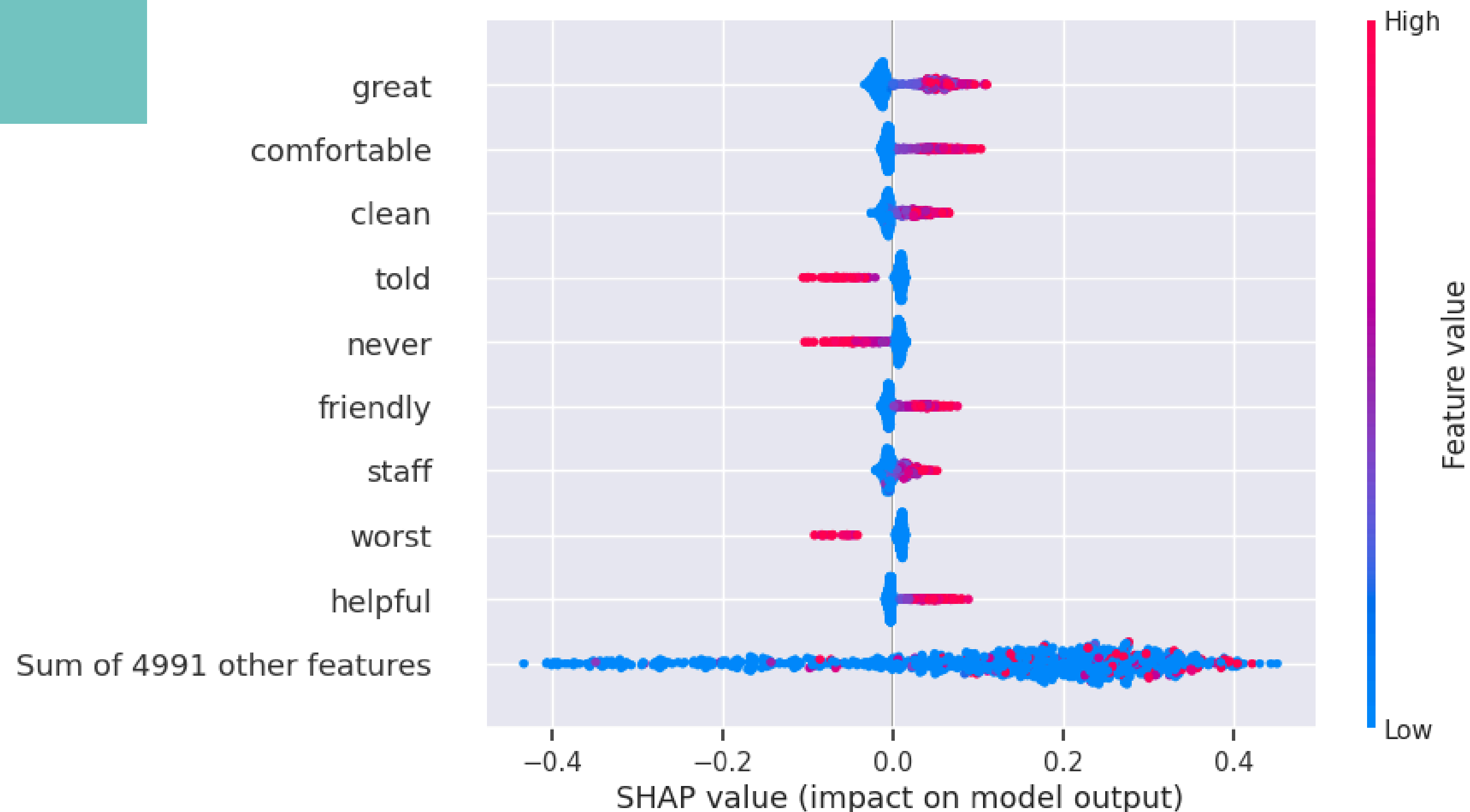
EXPLAINABLE AI - GLOBAL FEATURE IMPORTANCE



- Random Forest's feature importance
- Evaluates contribution of features to model's predictions
- Do not give explanation on polarity of words

EXPLAINABLE AI - SHAP

Global explanation

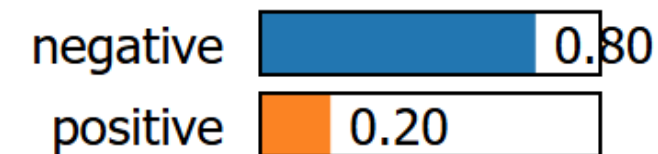


- Similar words of Random Forest's feature importance
- Here, we have an hint on polarity:
 - **"great"**: positive SHAP values suggest that when this word appears, the model tends to predict a more positive outcome
 - **"worst"**: negative SHAP values suggest that the presence of this word is associated with less positive outcomes (e.g., a negative review).

EXPLAINABLE AI - LIME

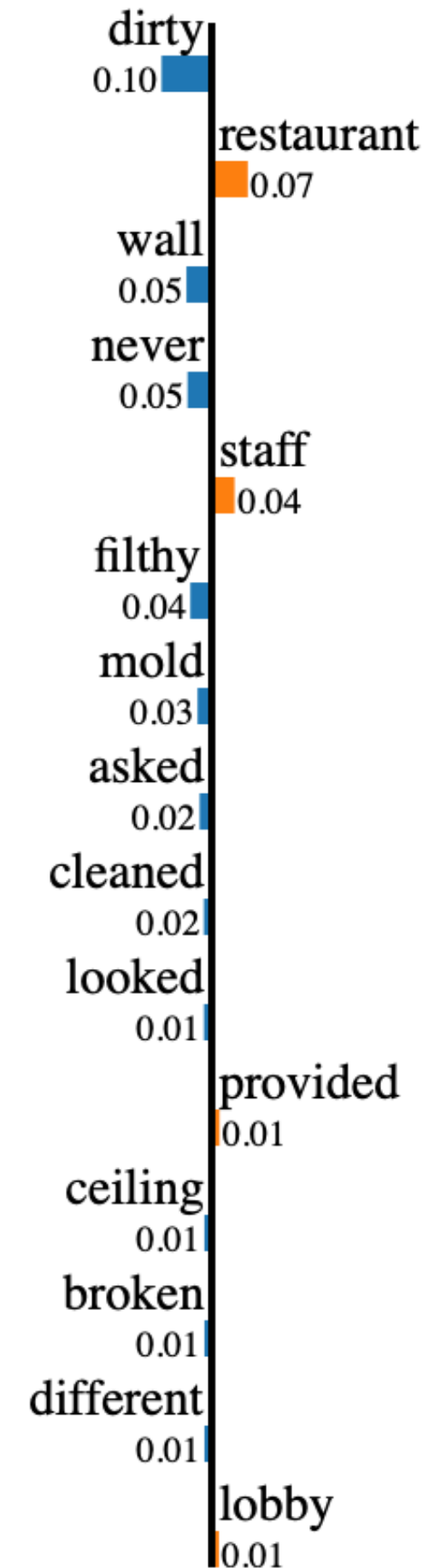
Local explanation

Prediction probabilities

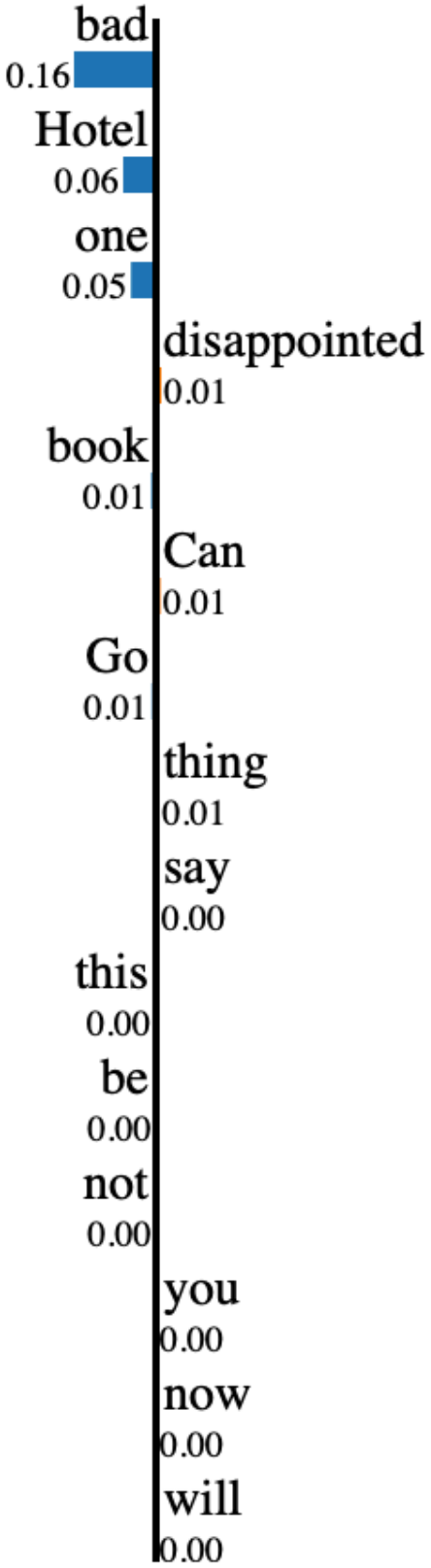


Moldy filthy rooms. (I looked at 6 different rooms). Heavy mold around air conditioner...asked for it to be cleaned twice (on 2 consecutive days) before switching rooms. Non responsive staff. NOT newly renovated. Half painted ceiling, broken air conditioning, dirty ceiling, rusty towel racks, ripped wall paper, poorly repaired walls. Property is poorly maintained. Housekeeping dumps dirty water buckets in parking lot. Litter on property is not cleaned regularly (at least not daily). I walked past the same empty bottle for 3 days straight. Pool never had towels available and lobby staff only provided towels when pushed. Only redeeming feature is the restaurant where you can get a decent meal anytime of the day. I cannot believe this is a Wyndham.

negative positive



EXPLAINABLE AI - LIME



RANDOM FOREST

Can not say one bad thing.
Go...book this Hotel now,
you will not be disappointed.

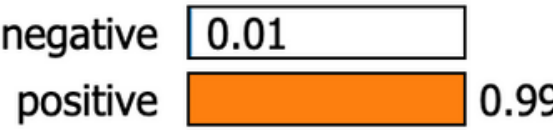
Prediction probabilities



FLAIR

Can not say one bad thing.
Go...book this Hotel now,
you will not be disappointed.

Prediction probabilities



Ground truth: 'positive'

THANK YOU

